

Lab Report No 1
Getting Started with Python
Artificial Intelligence



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Lab Task No 8:

Solution:

Brief description (3-5 lines)
Write a Python program to check if the given number is a Disarium Number. A number is said to be the Disarium number when the sum of its digit raised to the power of their respective positions becomes equal to the number itself. For example, 175 is a Disarium number as follows: $1^1 + 7^2 + 5^3 = 1 + 49 + 125 = 175$
The code
<pre>def is_disarium_number(number): # Convert the number to a string to easily iterate through its digits num_str = str(number) # Calculate the sum of digits raised to their respective positions disarium_sum = sum(int(digit) ** (index + 1) for index, digit in enumerate(num_str)) # Check if the sum equals the original number return disarium_sum == number # Test the function with some example numbers def test_disarium_numbers(): # Test cases test_numbers = [175, 89, 135, 518, 100, 200] for num in test_numbers: result = is_disarium_number(num) print(f'{num} is {'a' if result else 'not a'} Disarium number') # Run the test function test_disarium_numbers()</pre>
The results (Screenshot)

Lab Task No 9:

Solution:

Brief description (3-5 lines)

Write a Python program to determine whether the given number is a Harshad Number. If a number is divisible by the sum of its digits, then it will be known as a Harshad Number.

For example:

The number 156 is divisible by the sum (12) of its digits (1, 5, 6)

Some Harshad numbers are 54, 120, etc.

The code

```
def is_harshad_number(number):
    # Convert the number to a string to easily iterate through its digits
    num_str = str(number)
    # Calculate the sum of digits
    digit_sum = sum(int(digit) for digit in num_str)
    # Check if the number is divisible by the sum of its digits
    return number % digit_sum == 0

def main():
    # Test the function with some examples
    test_numbers = [54, 120, 156, 18, 10, 22, 100, 63]
    print("Harshad Number Checker")
    print("-" * 25)

    for num in test_numbers:
        result = is_harshad_number(num)
        print(f'{num} is {' ' if result else ' not '} a Harshad Number')
        if result:
            print(f' Digit Sum: {sum(int(digit) for digit in str(num))}')
        print()

# Allow the script to be run directly or imported as a module
if __name__ == "__main__":
    main()
```

The results (Screenshot)

```
Harshad Number Checker
-----
54 is a Harshad Number
  Digit Sum: 9

120 is a Harshad Number
  Digit Sum: 3

156 is a Harshad Number
  Digit Sum: 12

18 is a Harshad Number
  Digit Sum: 9

10 is a Harshad Number
  Digit Sum: 1

22 is not a Harshad Number

100 is a Harshad Number
  Digit Sum: 1

63 is a Harshad Number
  Digit Sum: 9
```

Conclusion

In conclusion, Harshad and disarium numbers are rare mathematical breeds, with Harshad numbers being divisible by their digit sum, and disarium numbers showing symmetry when raised to successive powers. These numbers demonstrate how numbers can be related through digits and mathematical relationships, allowing mathematicians and number theory enthusiasts to understand elegant patterns hidden in numerical sequences. These classifications have given glory to mathematicians and number theory enthusiasts.